

Classification of Clinical Application Modalities

Overview: We classified the predictive models according to their clinical application modalities, encompassing nomogram-based tools, point-based risk scores, digital implementation tools, and algorithm-only models.

Category 1 — Nomogram-Based Tools

Definition: Static or dynamic graphical tools that map predictor values to risk probability via point accumulation. Combines visual interpretability with quantitative precision, suitable for low-tech clinical settings (e.g., bedside, outpatient clinics).

Key Features: Includes “nomogram” in title/abstract; provides calibrated point scales; outputs probability (not just categorical risk).

Category 2 — Point-Based Risk Scores

Definition: Manual, integer-weighted scoring systems for rapid, resource-efficient risk stratification. Requires no electronic devices, ideal for time-sensitive or resource-limited scenarios.

Key Features: Keywords: “score” , “risk index” , “clinical rule” ; requires manual point addition; outputs categorical (low/medium/high) or numerical risk.

Category 3 — Digital Implementation Tools

Definition: Digital tools (standalone or integrated) that automate risk calculation via user input. Reduces human error and supports personalized risk visualization (e.g., risk curves, management recommendations).

Key Features: Keywords: “calculator” , “web-based tool” , “mobile app” , “EHR-integrated” ; clear user input interface; automated computation (no manual scoring).

Category 4 — Algorithm-Only Models

Definition: Predictive models reported solely as algorithms (e.g., ML, AI, statistical models) without clinical implementation tools. Focuses on methodological innovation rather than direct bedside application.

Key Features: Keywords: “machine learning” , “AI” , “statistical model” ; no score/nomogram/calculator; only algorithm description + performance metrics.